

# grna assignment results

2026-01-10

outline

- methods
- datasets (real and simulated)
- results: simulation ground truth analysis and real data

## 1. Methods and datasets

For this part of the project, we are assessing five methods (two of which are versions of sceptre) on two real datasets and one simulated dataset.

The methods:

1. perpty's `assign_mixture_model()`: Bayesian Poisson-Gaussian mixture model fit via MCMC with CUDA+JAX support
2. crispat's `ga_poisson_gauss()`: a Poisson-Gaussian mixture fit where the MAP estimates are obtained from a variational procedure
3. cleanser: Bayesian NB-NB mixture fit via MCMC with stan (their `direct-capture` version)
4. sceptre's mixture model: latent variable Poisson GLM fit via EM variant
5. new sceptre-pipeline

The datasets:

- a. Gasperini 2019 (high moi)
- b. Replogle 2022 (low moi)
- c. simulations

### 1a. Simulated data

We obtained a simulated low MOI dataset in the following way:

Inputs:

- a gRNA expression vector from Replogle, denoted  $Y$
- cell covariates  $z_i$ : in this case, the actual `grna_n_nonzero` and `grna_n_umis` from the sceptre covariate dataframe
- the probability a simulated cell is perturbed, denoted  $\varphi$
- intercept adjustments  $\gamma_0$  and  $\gamma_1$  for unperturbed and perturbed cells, respectively, and a new NB dispersion parameter  $\theta_0$

Model and simulate:

- do a NB regression of  $Y$  on the covariates  $Z$  to get their effects.

- draw a new  $Y^*$  from a NB mixture, with each cell's mean being determined by the linear predictor from the NB regression, except also with the intercept adjustments. The overdispersion is  $\theta_0$  no matter the perturbation status

More precisely, let  $Y = (Y_1, \dots, Y_n) \in \mathbb{N}^n$  represent the observed UMI counts for our chosen actual gRNA. For each cell  $i$ , let  $z_i$  give our cell covariates, and collect these into a matrix  $Z \in \mathbb{R}^{n \times p}$ . Do a NB regression of  $Y$  on  $Z$  to get  $\hat{\beta}$ . If we just wanted to simulate from this model, we could do  $Y_i^* | z \stackrel{\perp\!\!\!\perp}{\sim} \text{NB}(e^{z^T \hat{\beta}}, \hat{\theta})$ .

But for our case we need to have some cells be perturbed, so we are sampling from a 2-component NB mixture. Let  $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(\varphi)$  be our perturbation indicators, so  $X_i = 1$  means cell  $i$  is perturbed.

Let

$$\log \mu_i^* = z_i^T \hat{\beta} + \gamma_0(1 - X_i) + \gamma_1 X_i$$

so the effects for each covariate are the same, but the intercept is adjusted to be either  $\hat{\beta}_0 + \gamma_0$  or  $\hat{\beta}_0 + \gamma_1$ , depending on which component we are in. We then simulate

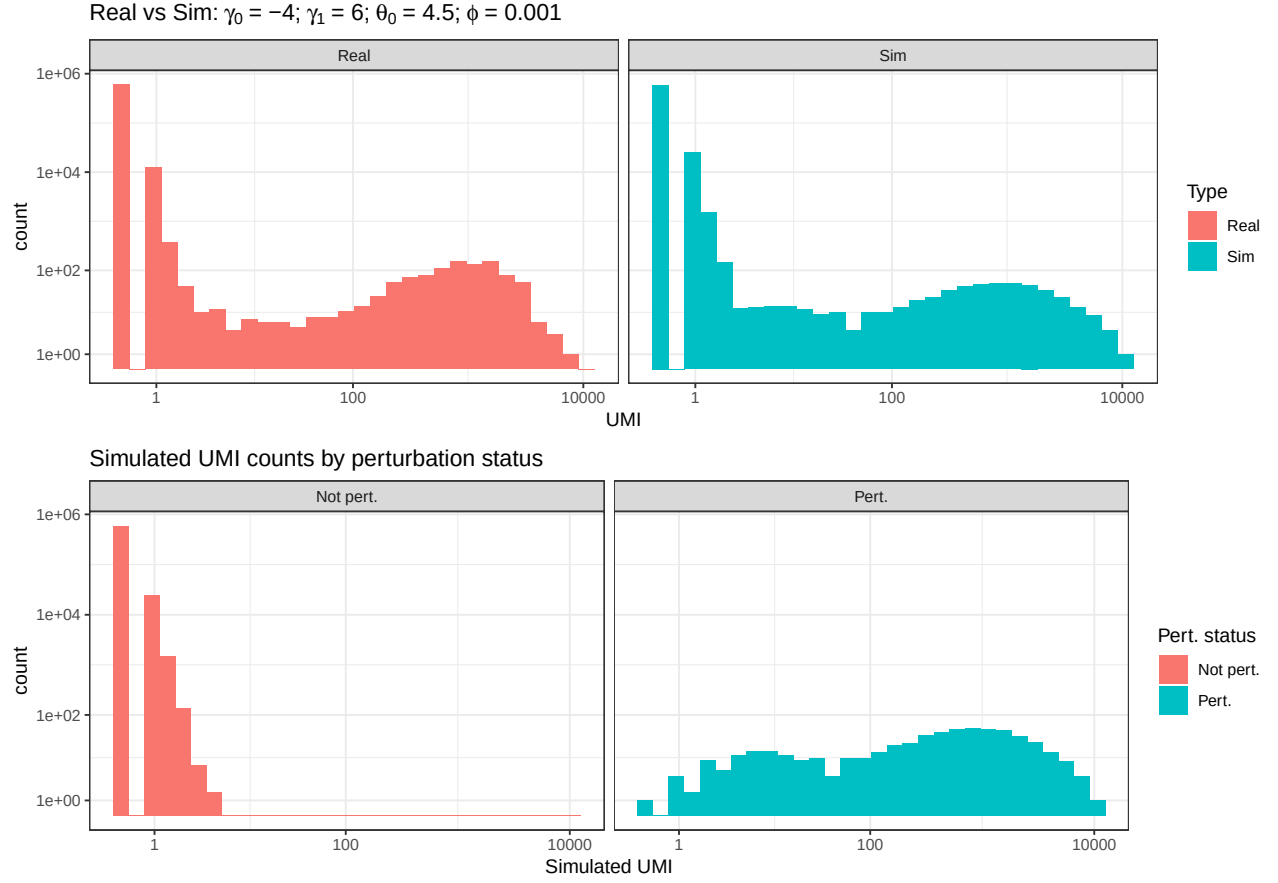
$$Y_i^* | z_i, X_i \stackrel{\perp\!\!\!\perp}{\sim} \text{NB}(\mu_i^*, \theta_0)$$

where  $\theta_0$  is our externally provided overdispersion.

We can't use  $\hat{\theta}$  here, the overdispersion as estimated by the NB regression, because that is marginal over the perturbation indicators. There is much less overdispersion when we know which component each cell is from. The larger  $\gamma_1 - \gamma_0$  is, the bigger the perturbation effect size.

For this paper, we use row 58 of the Replogle-rd7 gRNA matrix, which has gRNA id 304\_ANAPC15\_P1P2\_ENSG00000110200. We regress the resulting UMI counts on  $\sim 1 + \log(\text{grna\_n\_nonzero}+1) + \log(\text{grna\_n\_umis}+1)$ .

Below is a comparison of the actual UMI counts and a simulated dataset:



## 2. Results

### a. simulated data: comparison to ground truth

We ran on the medium dataset due to cleanser's memory usage.

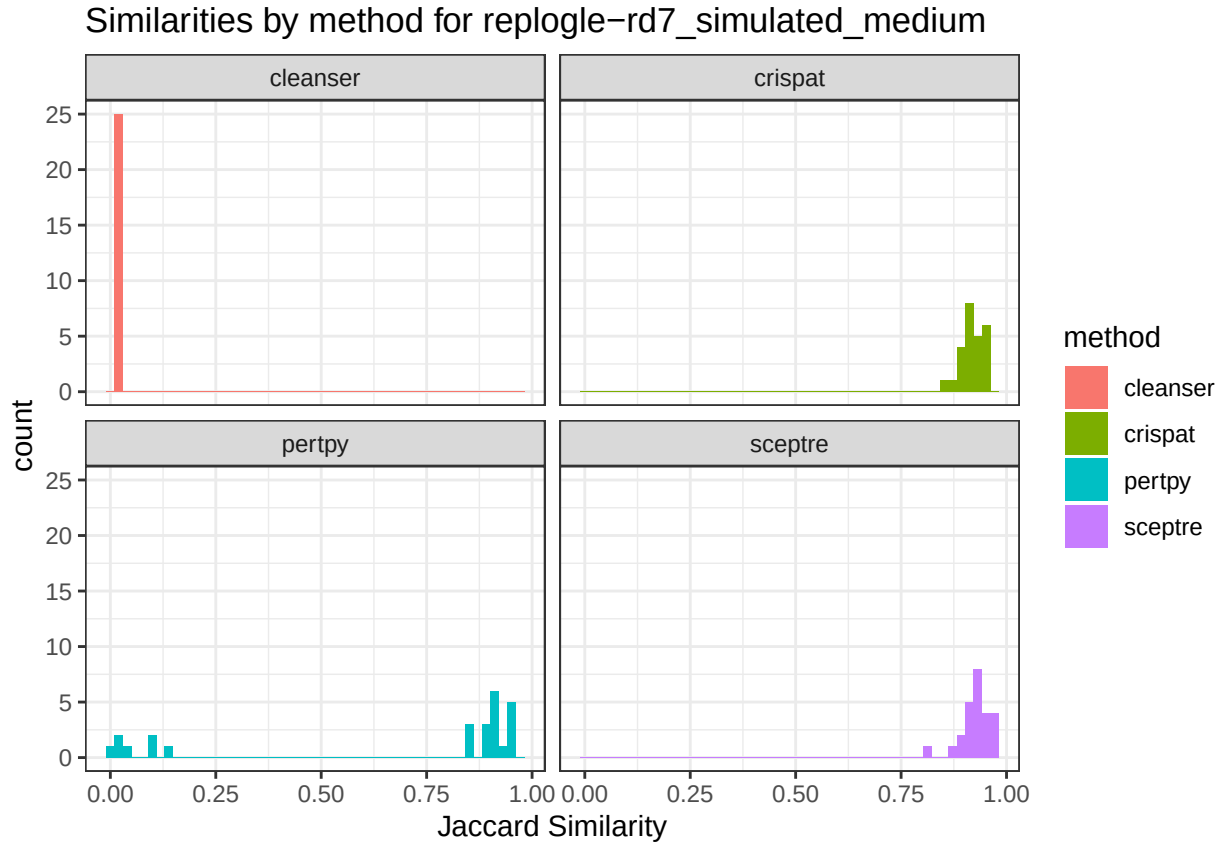
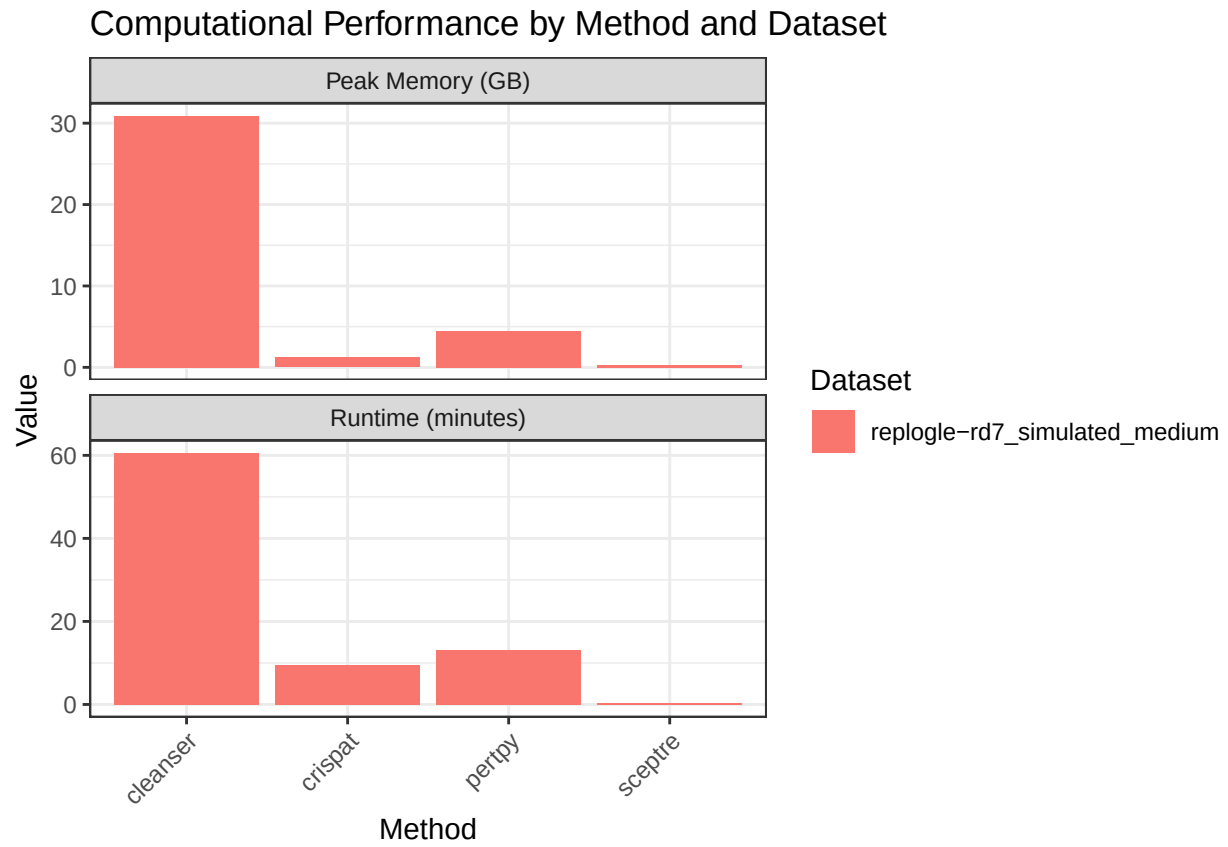
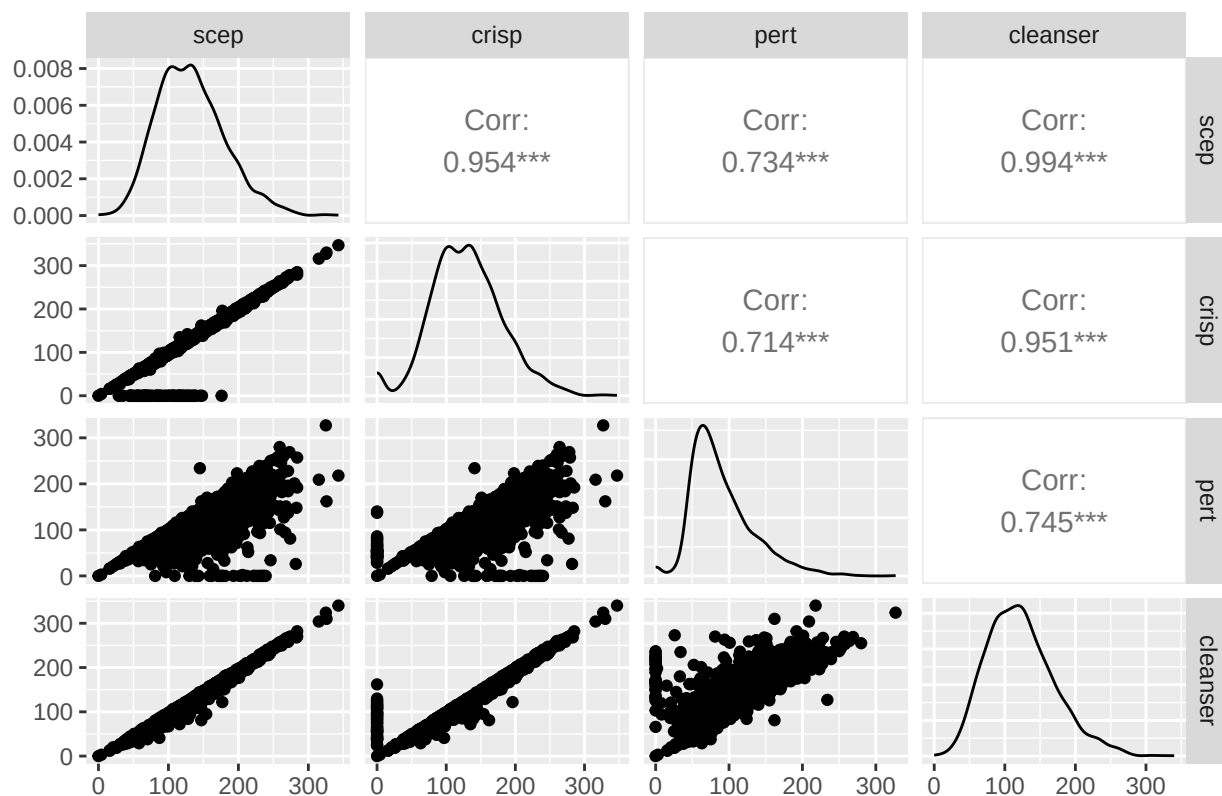
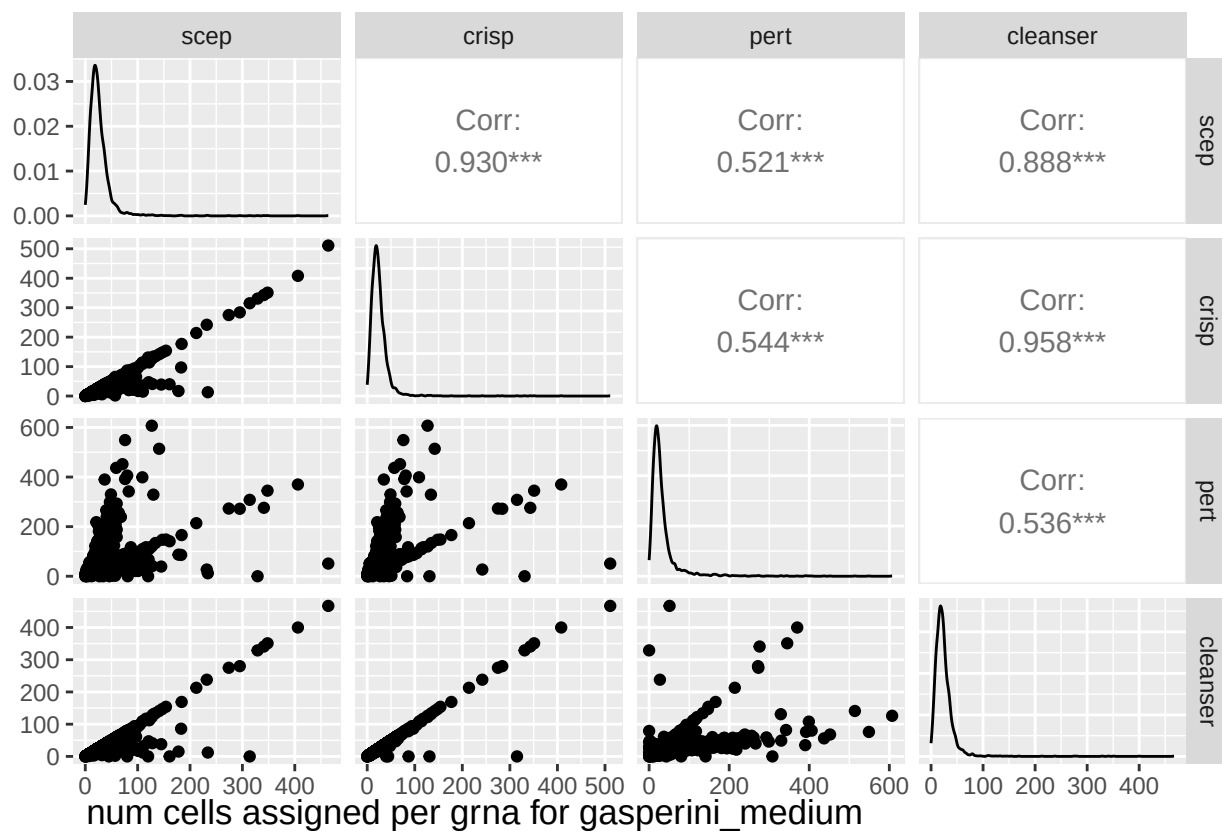


Figure 1: For each of the 25 iid simulated datasets, we get the similarity between the true perturbation status indicators and the estimated perturbation status. Cleanser totally fails; pertpy recovers some (if more MCMC samples are taken, pertpy does better); and sceptre and crispat nearly perfectly recover the ground truth.



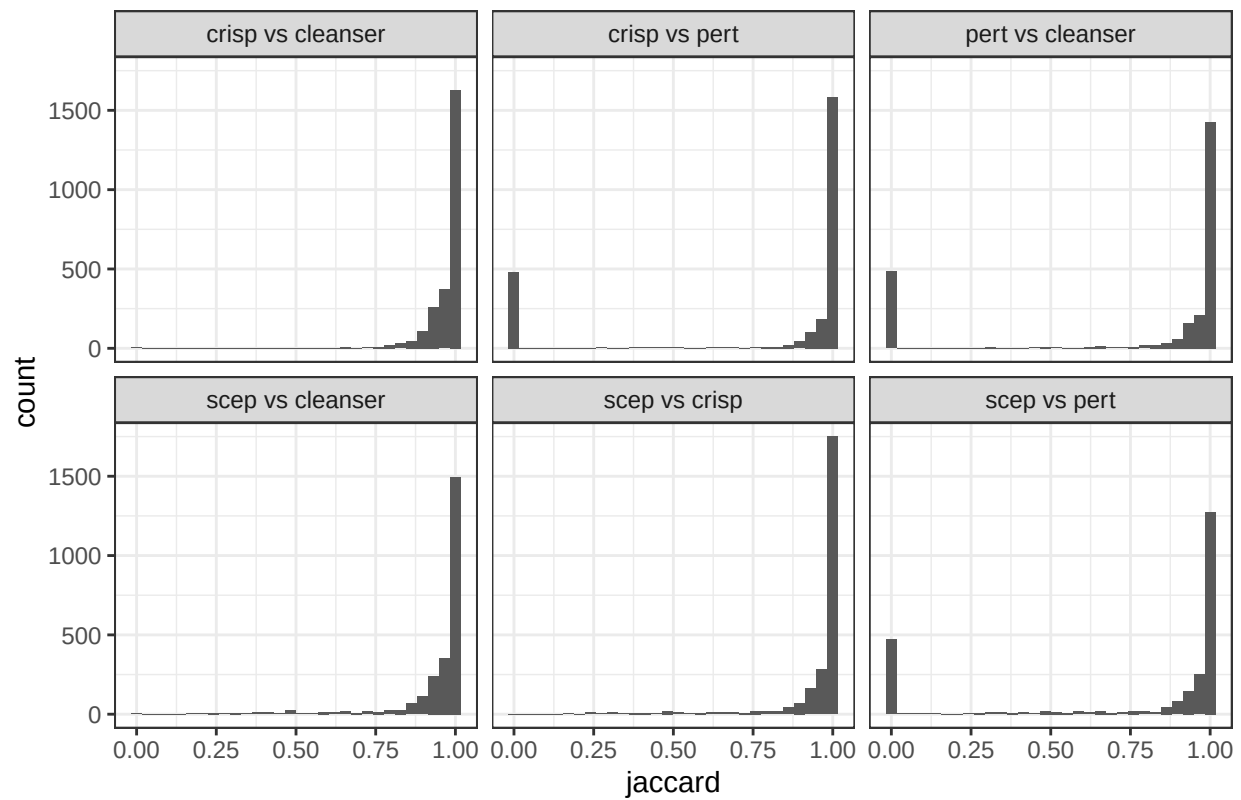
## 2. real data

num cells assigned per grna for replogle-rd7\_medium



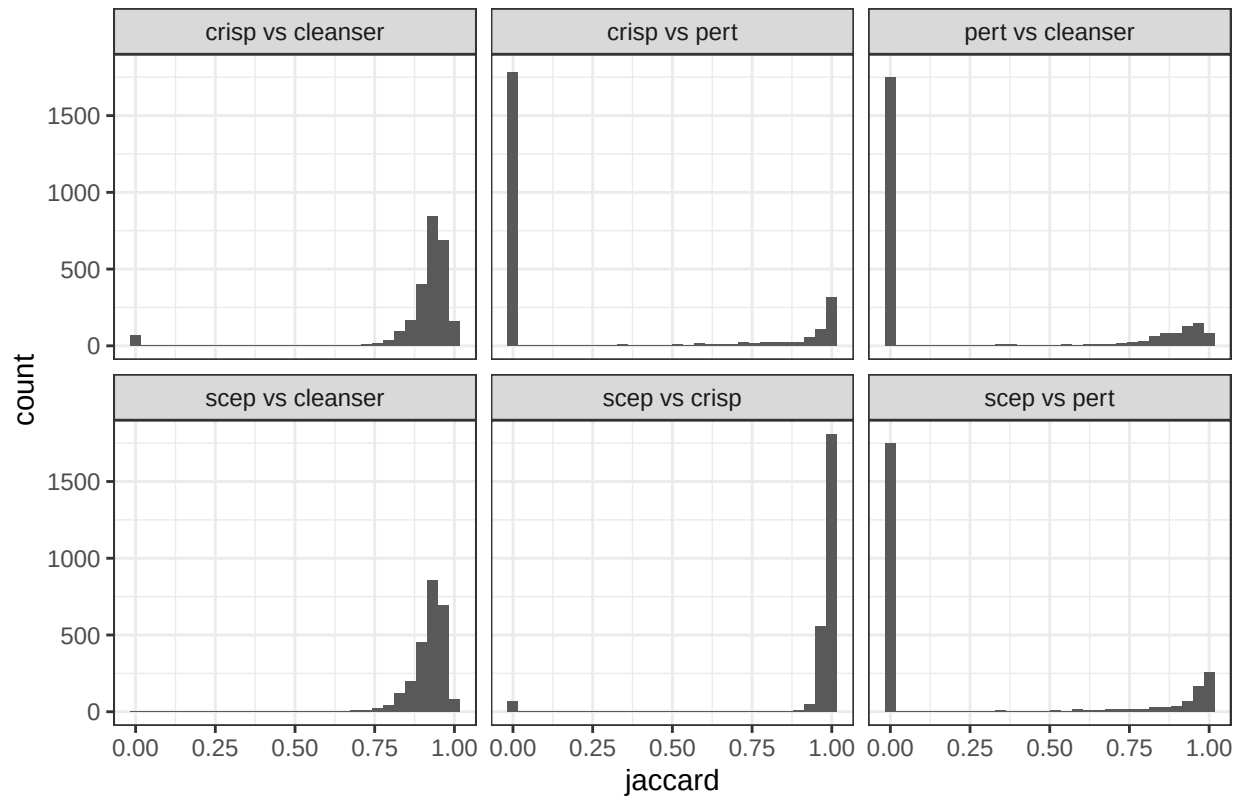
```
## Warning: Removed 13 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

### grna assignment similarities for replogle-rd7\_medium

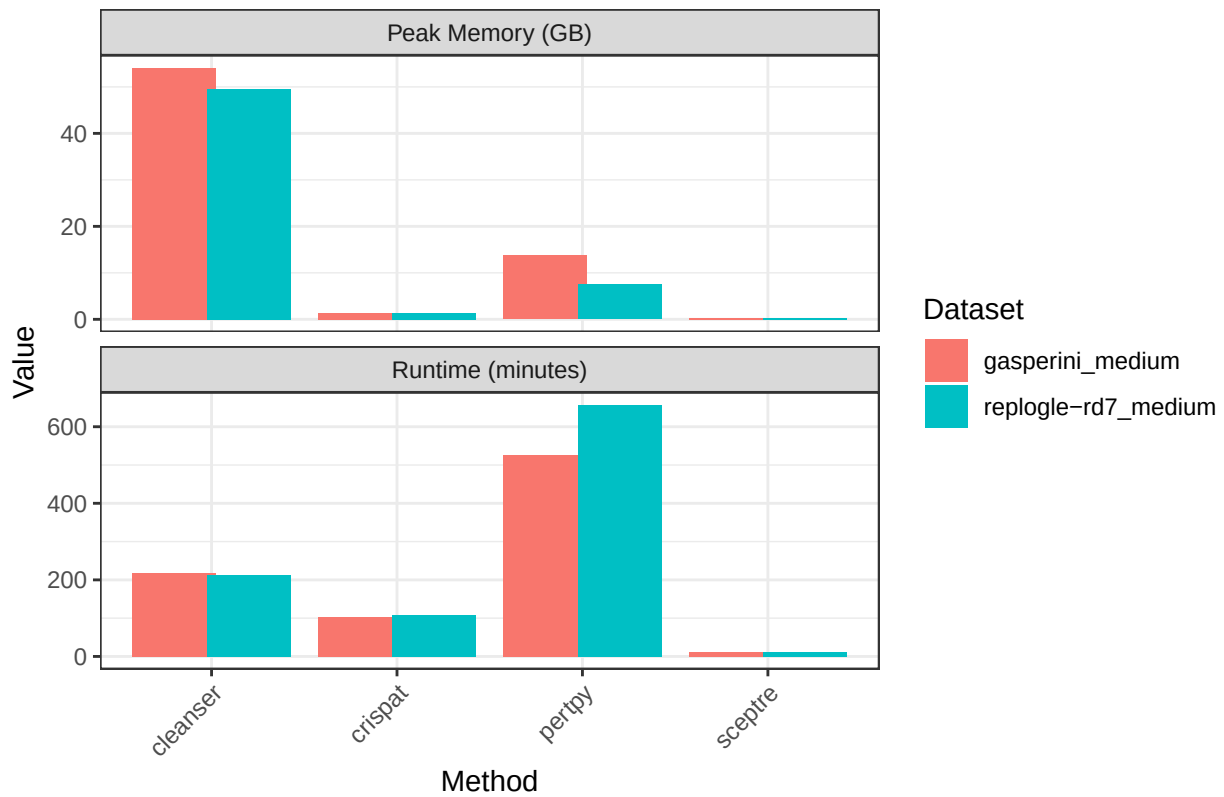


```
## Warning: Removed 6 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

grna assignment similarities for gasperini\_medium



Computational Performance by Method and Dataset





## Sceptre-Pipeline Performance by Dataset

